

HW2 Critical Thinking Lab: Questions 3 and 4 (Graduate Track)

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Section	Monday and Wednesday
Page tested	hw2/index.html with styles.css and print.css
Main page (Codd)	https://codd.cs.gsu.edu/~anuni1/HW2/hw2/
Q4 defect copy (Codd)	https://codd.cs.gsu.edu/~anuni1/HW2/q4-defect/
Q4 fixed copy (Codd)	https://codd.cs.gsu.edu/~anuni1/HW2/q4-fixed/
GitHub repo	https://github.com/nuniaishwaryachowdary17/Homework02

About these screenshots

Everything was captured in one browser: Chromium in headless mode, driven by Playwright. Reduced motion was on, so the entrance animation couldn't shift any boxes while I measured. The viewports are exactly 700 and 480 CSS px wide. The machine had substitute fonts for Verdana and Tahoma, so heights can land a few pixels off on another computer. The colored outlines, labels and red center lines in the annotated shots were drawn on top by the test script. They are not in the submitted page. The two desktop screenshots on the last page were taken in my own Chrome with the page address visible.

Question 3: Architecture defense

Measurement	700 px	480 px
Horizontal overflow (scrollWidth minus clientWidth)	0	0
#content: display and grid-template-columns	grid / 672px	grid / 452px
Named areas (computed)	"rating" "reviews" "overview" "count"	same
#rating, width x height	672 x 83	452 x 67.8
#reviews, width x height	672 x 941.9	452 x 968.1
#overview, width x height	672 x 745.3	452 x 769.3
#count, width x height	672 x 22	452 x 22
Left edge (x) of all four areas	14, all equal	14, all equal
#reviews flex: direction / justify / align	column / space-between / flex-start	column / space-between / flex-start
.column width x height (left, right)	645.1 x 463 ; 645.1 x 452	433.9 x 475 ; 433.9 x 475
.critic byline: display / wrap / align / gap	flex / wrap / center / 5px	same
Bylines where icon center = text center (of 8)	8 of 8	8 of 8
Bylines that wrapped (of 8)	0	0
Same bylines with display: block (flex removed): centers equal / largest offset	0 of 8 / 18.5 px	0 of 8 / 18.5 px

One extra check at 1024 px, above the breakpoint: the tracks are 550px and 250px. The overview starts at y=131, the same as the rating, and ends at y=948.3, the same as the bottom of the reviews. One grid item spans two rows.

Key: pink is the rating area, blue the reviews, green the overview, orange the count bar, each labeled with its measured width x height. Purple boxes are the 8 flex byline rows, and the red line through each one is the shared vertical center of icon and text. Clean copies are q3-700-clean.png and q3-480-clean.png.

Viewport 700 px. Page: hw2/index.html.
Overflow 0. Tracks 672px.

Viewport 480 px. Page:
hw2/index.html. Overflow 0.
Tracks 452px.

Question 3: what belongs to Grid and what belongs to Flexbox

Code under test (styles.css lines 51-61, 105-111 and 158-180)

```
@supports (display: grid) {
  #content {
    display: grid;
    grid-template-columns: 1fr 250px;
    grid-template-rows: auto 1fr auto;
    grid-template-areas:
      "rating overview"
      "reviews overview"
      "count count";
  }
}
.critic {
  display: flex;
  flex-wrap: wrap;
  align-items: center;
  gap: 5px;
  margin: 10px 0 0;
}
/* 700px and below: one column, and the two review columns stack */
@media (max-width: 700px) {
  #content { width: auto; margin: 0 10px 60px; }
  @supports (display: grid) {
    #content {
      grid-template-columns: 1fr;
      grid-template-areas: "rating" "reviews" "overview" "count";
    }
  }
  #overview img { width: 100%; max-width: 250px; }
  #reviews { flex-direction: column; }
  #reviews .column { width: 100%; }
}
```

Grid runs the page regions

The four regions depend on each other in two directions. Above 700 px the overview sits beside the rating and the reviews and spans both rows, and the count bar runs underneath all of it. Named areas let me write that relationship straight into the template. At 1024 px the overview begins where the rating begins (y=131) and ends where the reviews end (y=948.3), which is one item covering two rows.

At 700 px I swap the template for a single column. That is four lines of CSS and no HTML edits. Every area then measures 672 px wide and starts at x=14, so the edges line up exactly. At 480 px the same template gives 452 px tracks and the same left edge. The source order never changes, so tab order and reading order match what you see on screen at every width. If a browser has no Grid support, @supports leaves #content as a plain block and the regions stack in source order, which looks just like the narrow layout.

Flexbox runs the byline and the review columns

An icon next to a name and publication is a one-direction row, and the page repeats it 8 times. With align-items: center the icon and the text share a center in 8 of 8 bylines at both widths. When I switched to display: block in the test script (only there, never in the page), 0 of 8 matched and the icon sat up to 18.5 px off, because an inline image sits on the text baseline. Grid would need an explicit template for a row that has no second direction. Flexbox does the job with one declaration.

The two review columns are flex too. #reviews switches from row to column at 700 px with one line. align-items: flex-start keeps each column at its natural height: 463 and 452 px at 700, and 475 and 475 px at 480. The columns are 645.1 px wide at 700 and 433.9 px at 480, inside the 672 and 452 px tracks, so nothing overflows.

What flex-wrap actually does

I narrowed the viewport one pixel at a time. At 210 px and wider no byline wraps. One wraps at 209 px, and 2 of 8 wrap from 205 px down. So flex-wrap is a safety net far below 480 px. It is not what holds the 700 and 480 px layouts together, since 0 of 8 wrapped at both widths. Grid does that.

Short version for the live defense: Grid for how regions relate to each other, because the sidebar spans rows and the count bar spans columns. Flexbox for spreading items along one line inside a region. Measured: overflow 0 at both widths, tracks of 672 and 452 px, byline centers equal in 8 of 8.

Question 4: Failure analysis

I picked a clipped review as the defect. It lives in a separate copy, q4-defect/. The original hw2/ folder, which every Question 3 screenshot came from, was left alone, and the repaired copy is q4-fixed/. Two declarations were added to .quote in q4-defect/styles.css, on lines 98 and 99:

```
.quote {
  display: flow-root;
  margin: 0;
  padding: 8px;
  background: #E1D697;
  border: 2px solid gray;
  font-weight: bold;
  height: 2.5em; /* DEFECT (intentional): sized from the short quotes */
  overflow: hidden; /* DEFECT (intentional): hides whatever does not fit */
}
```

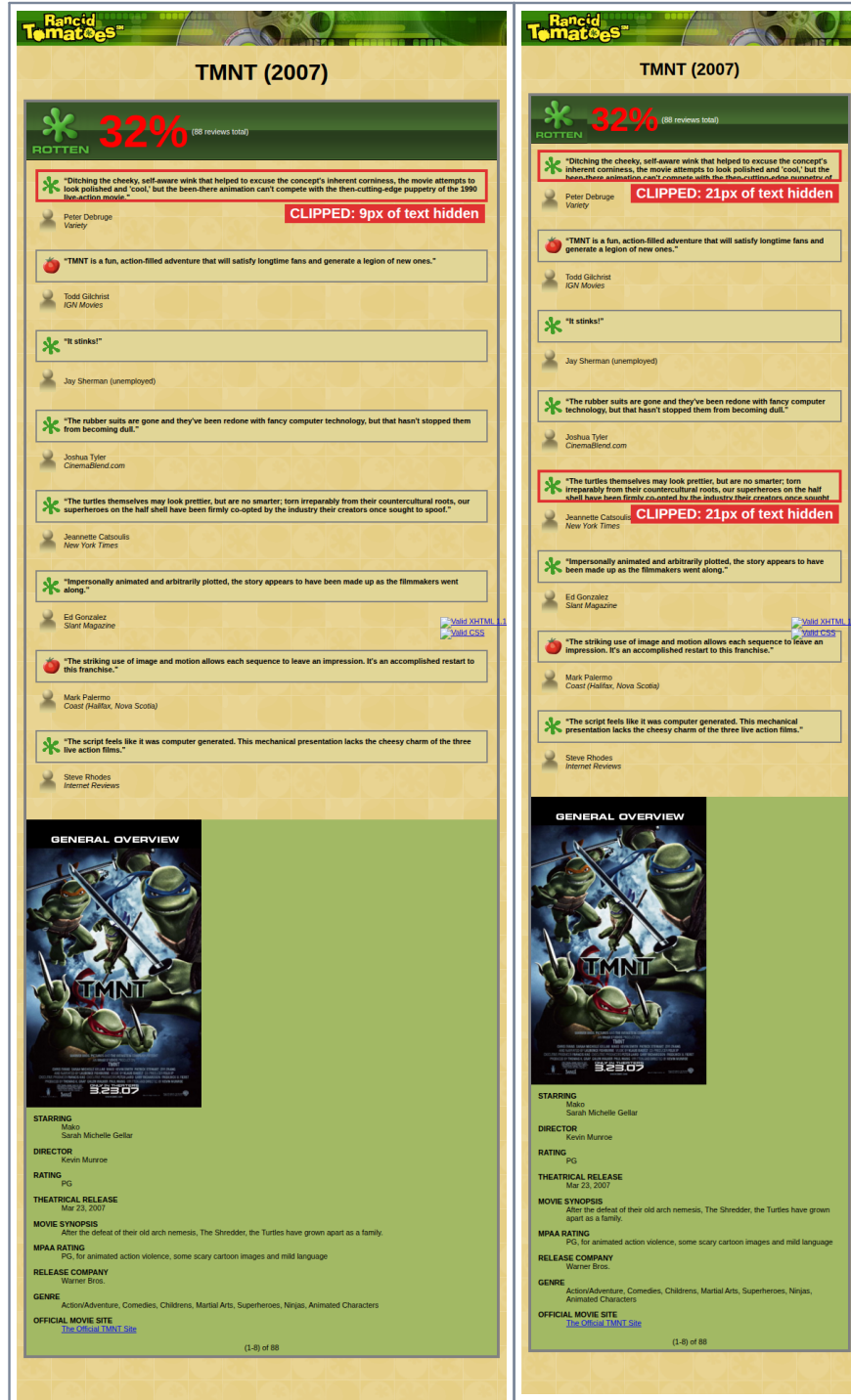
The scenario is "make every review card the same height", sized from the short two-line quotes. 2.5em works out to 26.7 px of content height.

Defect state: how many reviews get clipped (scrollHeight > clientHeight)

Viewport	Clipped	Which ones	Text lost	Page height, original to defect
1024 px	7 of 8	all except #3 ("It stinks!")	not recorded	1034 to 1034
700 px	1 of 8	Peter Debruge	the end of the quote, "live-action movie."	1987 to 1989
480 px	2 of 8	Peter Debruge and Jeannette Catsoulas	last two lines of each	2013 to 1980

The count moves with the width (7, 1, 2) because the line wrapping moves with it, so a check at a single width would miss the problem. The page height changes too. At 480 px it ends up 33 px shorter, since the hidden lines no longer take up room. Nothing on screen tells the reader that text is missing.

Question 4: defect state (screenshots)



DEFECT, viewport 700 px. Page: q4-defect/index.html. Red box = clipped quote.

DEFECT, viewport 480 px. Page: q4-defect/index.html. Two clipped quotes.

Question 4: diagnosis from browser output and DevTools

Step 1: the symptom. At 480 px two quote boxes end mid-sentence, and at 700 px one does, with the last line cut off at the bottom border. Horizontal overflow is 0 at both widths, so this is not a width problem.

Step 2: the Styles pane. I inspected the first .quote through the DevTools protocol (CSS.getMatchedStylesForNode). The author rules that apply:

```
.quote { display: flow-root; } /* styles.css:92 */
.quote { padding: 8px; }      /* styles.css:94 */
.quote { border: 2px solid gray; } /* styles.css:96 */
.quote { font-weight: bold; }  /* styles.css:97 */
.quote { height: 2.5em; }      /* styles.css:98 */
.quote { overflow: hidden; }   /* styles.css:99 */
```

Step 3: the Computed pane (CSS.getComputedStyleForNode) at 700 px:

```
height: 26.6562px  overflow-x: hidden  overflow-y: hidden
box-sizing: content-box  padding-top: 8px  padding-bottom: 8px
border-top-width: 2px  display: flow-root
```

Step 4: the console. For all 8 quotes, [clientHeight, scrollHeight]. A quote is cut wherever scrollHeight is bigger than clientHeight:

```
700px: [[43,52],[43,43],[43,43],[43,43],[43,43],[43,43],[43,43],[43,43]]
480px: [[43,64],[43,43],[43,43],[43,43],[43,64],[43,43],[43,43],[43,43]]
clipped at 700px: #1.  clipped at 480px: #1 and #5.
```

To reproduce in your own browser:

```
[...document.querySelectorAll(".quote")].map(q=>[q.clientHeight,q.scrollHeight])
```

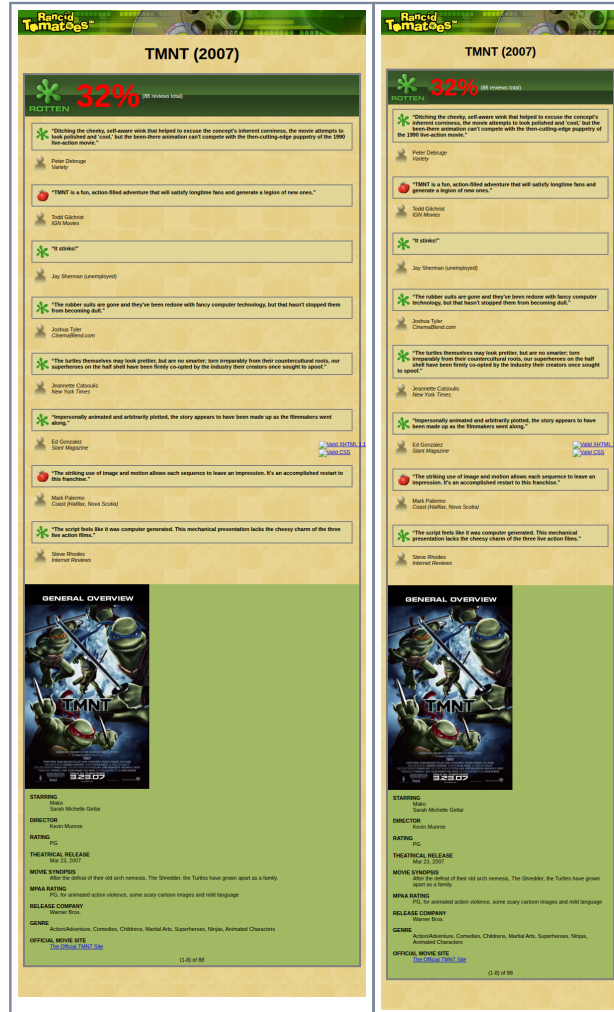
Root cause

height: 2.5em pins the box at 26.7 px of content height (clientHeight 43 once the 8px padding is counted). That fits a two-line quote and nothing longer. overflow: hidden then turns the extra lines into silent loss instead of letting the box grow. Put plainly, it is a fixed height on content of unknown length. How many quotes get hit depends on the width (1 at 700 px, 2 at 480 px, 7 at 1024 px) because the number of wrapped lines depends on the width.

The fix

Delete both lines. That is the only difference between the two copies. If equal-height cards were really the goal, min-height (it only grows) or a Grid row that stretches the cards would do it. height plus overflow: hidden never will.

Question 4: corrected state



CORRECTED, viewport 700 px.
Page: q4-fixed/index.html. 0
clipped.

CORRECTED,
viewport 480 px.
Page:
q4-fixed/index.html. 0
clipped.

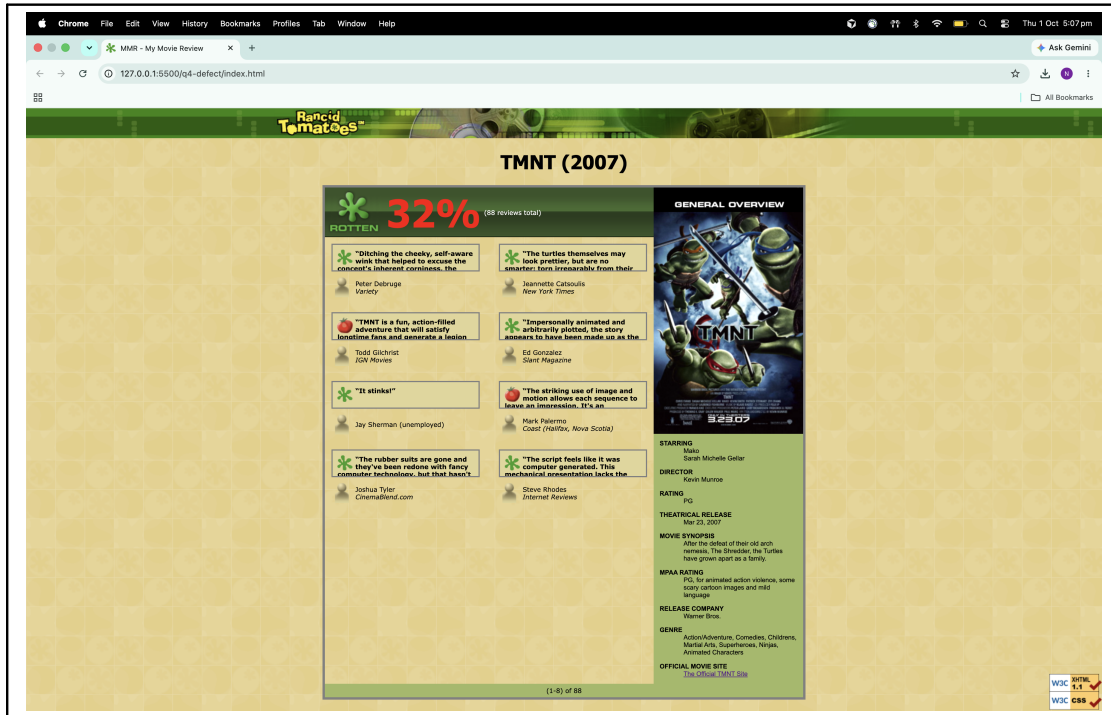
Check on q4-fixed	700 px	480 px
Clipped reviews (scrollHeight > clientHeight)	0	0
Horizontal overflow	0	0
Page height: original / defect / corrected	1987 / 1989 / 1987	2013 / 1980 / 2013
Pixel difference, corrected vs original screenshot	none (identical)	none (identical)

The corrected copy is byte-identical to the original hw2/ files, and the Question 3 screenshots came from hw2/, so the original evidence is untouched. The defect shots (q4-defect-*.png) and the corrected shots (q4-fixed-*.png) are separate files.

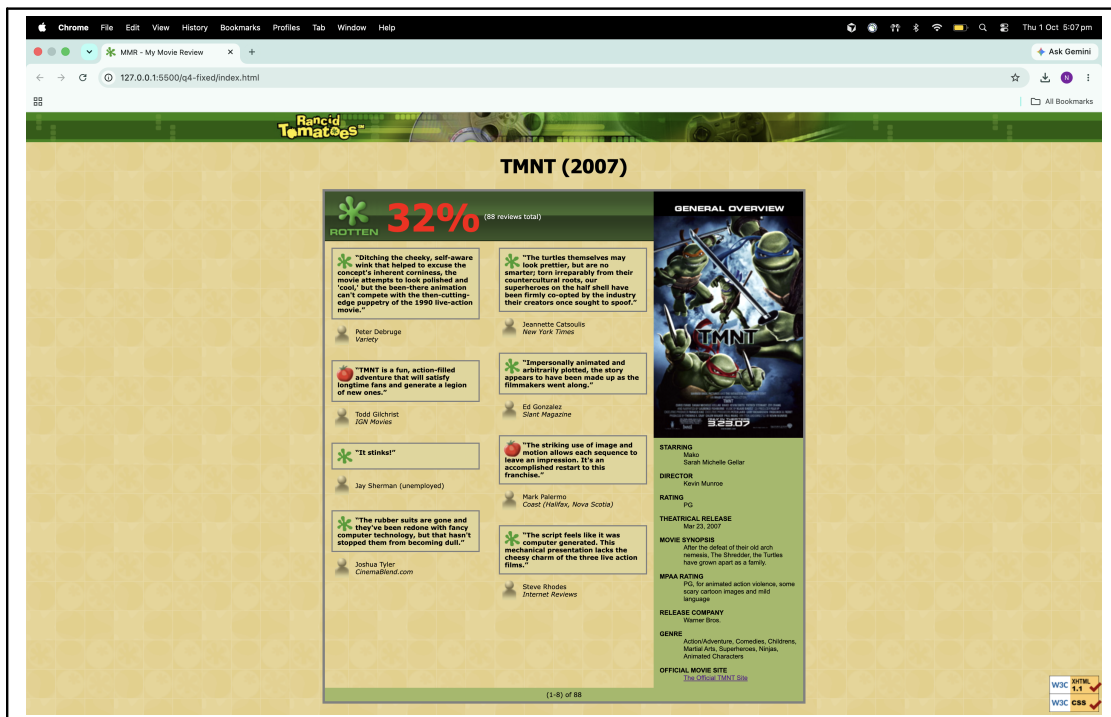
Short version for the live defense: a fixed height with overflow hidden, sized from short content, hides longer content. DevTools showed it as height 2.5em plus overflow hidden on .quote (lines 98 and 99), a computed height of 26.7 px, and scrollHeight above clientHeight on the clipped quotes. Removing the two lines gives 0 clipped at 700, 480 and 1024 px.

Question 4: defect and corrected state in my own browser (address bar visible)

Chrome, desktop window, page not resized (the screenshots are 3600 px wide on a 2x display, so the window is about 1800 CSS px). Both were served from the same folder on my machine. The top one is q4-defect, the bottom one is q4-fixed.



DEFECT, 127.0.0.1:5500/q4-defect/index.html. Seven of the eight quotes are cut off at the bottom border. Only "It stinks!" fits. This matches the 7 of 8 count in the defect table.



CORRECTED, 127.0.0.1:5500/q4-fixed/index.html. Every quote shows in full and the boxes grow to fit.